SMACC

Sleep Manipulation And Communication Clickything — manual

SMACC

None

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1. SMACC

Sleep Manipulation And Communication Clickything (SMACC) is a Windows desktop app for running sleep and dream studies — presenting cues to a sleeping participant, communicating with them, marking events on the EEG, and collecting dream reports.

In a dream-engineering session the experimenter has to deliver a precisely-timed cue to a sleeping participant, mark each event so it lines up with the EEG, talk with the sleeper, and collect a dream report — all from a dark control room, late at night, on whatever hardware the lab has. SMACC is the clickable control surface for that work: one window per job, glanceable in the dark, with a volume safety cap and explicit device routing so a misclick can't blast or mislead a sleeping participant.

Download SMACC **Installation guide**

The button downloads the installer (`SMACC-Setup.exe`) for the latest version, which installs per-user (no admin rights) and runs on 64-bit Windows 10 or later. Prefer a single file with no install? A portable `SMACC.exe` ships with every release — see Installation, which also covers the Windows SmartScreen warning on the unsigned download and how to get older versions.

Working offline? Download these docs as a single PDF.

1.1 What SMACC does

- Trigger **audio cues** (and design them in-app with the Audio Cue Designer)
- Trigger **visual cues** on a BlinkStick or Philips Hue light
- Play masking **background noise**
- Run **biocalibrations** as timed, marked tasks
- Record **dream reports** and administer **surveys**
- **Talk, listen, and type** with the participant over an intercom
- Mark events with **EEG port codes** over LSL or a hardware TTL trigger
- Review and score recordings in the **EEG Annotator**
- Save a detailed **event log** for every run

1.2 Get started

Getting started

- Installation — download and run SMACC.
- SMACC files — create and reuse a study's configuration (`.smacc`).

Running a session

- Overview — the Launcher, the Session window, and the tools.
- Audio cues · Visual cues · Biocals · Dream reports & surveys · Intercom & chat · Markers & port codes

Devices, volume & timing

- Audio routing · Compatible devices · Volume & latency

After the night

- EEG Annotator — review and score recorded EEG.
- Troubleshooting · Reference · Contributing

1.3 Used by

SMACC is used in dream engineering research, including by:

- Ken Paller's Cognitive Neuroscience Lab at Northwestern University
- Torres-Platas et al. (2026). Intentional lucid dreaming with a transformative learning agenda. *Research Square* doi:10.21203/rs.3.rs-8745420/v1
- Konkoly et al. (2026). Using real-time reporting to investigate visual experiences in dreams. *J Cogn Neurosci* doi:10.1162/jocn.a.107
- Morris et al. (2026). Inducing lucid dreaming based on a contemplative practice of compassion. *Brain Sci* doi:10.3390/brainsci16030315
- Konkoly et al. (2026). Creative problem-solving after experimentally provoking dreams of unsolved puzzles during REM sleep. *Neurosci Conscious* doi:10.1093/nc/niaf067
- Konkoly et al. (2025). Investigating dreams by strategically presenting sounds during REM sleep to reactivate waking experiences. *Neuropsychologia* doi:10.1016/j.neuropsychologia.2025.109229
- Morris et al. (2025). Lucid dreaming of a prior virtual-reality experience with ego-transcendent qualities: A proof-of-concept study. *Neurosci Conscious* doi:10.1093/nc/niaf017
- Mundt et al. (2024). Treating narcolepsy-related nightmares with cognitive behavioural therapy and targeted lucidity reactivation: A pilot study. *J Sleep Res* doi:10.1111/jsr.14384
- Wolk et al. (2024). Lucid dreams from reactivating mindfulness during REM sleep: A pilot study. *Int J Dream Res* doi:10.11588/ijodr.2024.2.98233
- Michelle Carr's Dream Engineering Lab at the University of Montreal and the Center for Advanced Research in Sleep Medicine
- Jafarzadeh Esfahani et al. (2024). Highly effective verified lucid dream induction using combined cognitive-sensory training and wearable EEG: A multi-centre study. *bioRxiv* doi:10.1101/2024.06.21.600133

SMACC is free software released under the GPL-3.0-or-later license.

2. Getting started

2.1 Installation

Download SMACC

The button above always downloads the installer for the **latest** version — double-click `SMACC-Setup.exe` and click through. The installer:

- installs SMACC **per-user** — no administrator rights or IT involvement needed;
- adds a **Start menu** entry (and, if you opt in, a desktop shortcut);
- associates `.smacc` **files**, so double-clicking one opens the **Start a session** dialog with it preselected;
- installs the **EEG Annotator** — the post-hoc EEG viewer/annotator. Included by default; uncheck it during setup to skip the heavyweight EEG component (which bundles the MNE library), or re-run the installer later to add or remove it;
- registers an uninstaller — remove SMACC from **Settings › Apps** like any other program. Uninstalling never touches your data: SMACC files, recordings, and logs under `~/SMACC` (or your data directories) all stay put. The uninstaller reminds you of this (and of the folder's location) when it finishes, so finding the folder afterwards is expected — delete it manually if you no longer need the data.

Installing a newer version over an existing one upgrades it in place, keeping whichever components you had chosen.

To get an older or specific version, browse the releases page: pick the release you want, open its *Assets* dropdown, and download from there. (The version switcher on this documentation site switches the *docs* only — older copies of SMACC itself come from the releases page.)

System requirements

SMACC runs on 64-bit Windows 10 or later.

Windows SmartScreen — "Windows protected your PC"

SMACC isn't code-signed yet, so when you run the installer Windows SmartScreen may show a blue **"Windows protected your PC"** box. This is expected for any new, unsigned program. To proceed anyway, click **More info**, then **Run anyway**.

Administrator privileges and the UAC prompt

For some features you will need to open SMACC with Administrator privileges (right-click the Start menu entry and select **Run as administrator**). Windows then shows a **User Account Control (UAC)** prompt asking whether to allow the app to make changes — click **Yes** to continue. For everyday use (audio cues, dream reports, LSL markers) you can run SMACC normally, without administrator rights.

2.1.1 Portable SMACC.exe (no install)

Every release also ships the same app as a single portable `SMACC.exe` — download it, double-click it, no installation. This suits USB-stick deployment, machines where nothing may be installed, and quick tests. The portable build skips the installer's conveniences (Start menu entry, uninstaller, the `.smacc` file association) — you can associate `.smacc` files from the Launcher's File menu instead (see below). The EEG Annotator is likewise available portable as `SMACC-EEG.exe` — put it **next to** `SMACC.exe` and the Launcher's *EEG Annotator* button lights up (it also runs standalone).

> **For IT departments**
>
> The default install is per-user (under `%LOCALAPPDATA%\Programs\SMACC`, no elevation). On managed machines where per-user installs are blocked — e.g. AppLocker policies on `%LOCALAPPDATA%` — run the same installer machine-wide instead: `SMACC-Setup.exe /ALLUSERS` (elevates, installs to Program Files).

2.1.2 Updating SMACC

From the Launcher, **File › Check for updates…** asks GitHub whether a newer release exists and, if one does, offers to open the download page in your browser — run the new installer and it upgrades the existing install in place. SMACC never checks on its own: lab machines are often offline, and studies usually pin one version for their whole run, so checking is always an explicit click. If you use the portable `SMACC.exe`, download the new one and replace your old copy.

2.1.3 After installing

By default SMACC stores everything under `~/SMACC`. To use a different location, set the environment variable `SMACC_DIRECTORY` to the directory you want; SMACC creates it and its subfolders on first run.

The rest of setup is covered on the relevant pages:

- **Your study configuration** lives in a portable SMACC file. SMACC seeds a `default.smacc` and opens it when you don't pick another, so it works out of the box. The installer associates `.smacc` files; with the portable build, enable or repair the association from **File › Associate .smacc files (Windows)** in the Launcher.
- **Audio cues** go in the data directory's `cues/` folder, alongside the seeded `demo-*` cues — see Audio cues.
- **Dream-report surveys** are managed from the Dream recording panel — see Dream reports & surveys.
- **A recording mic:** to record dreams, bind your mic to **Bedroom mic 1** in the **Devices** window (the **Panels** column) — see Audio routing.

2.2 SMACC files

A **SMACC file** (`.smacc`) is the study's configuration: cue files, volumes, noise, visual cues, biocals, survey presets, event markers, the display choices that apply to a session (always-on-top and log-preview levels), and the **data directory** where runs are written. It plays the same role for a SMACC study that a montage file plays for an EEG setup — configure it once, then reuse it session after session.

A SMACC file is *not* a session file. It is not tied to any particular run: opening one **starts a new session** with that configuration, and you can start as many sessions from the same file as you like. Sessions never rewrite the SMACC file they were started from — what happened in a run is recorded in that run's session log, not in the SMACC file.

It is a single, portable file (plain YAML you can read and edit). Cue/noise files and the data directory are stored **relative** to the SMACC file when they sit beside it — so a self-contained study folder stays valid if you move, copy, or zip it — and **absolute** when they point elsewhere. For the exact on-disk format, see the SMACC file reference.

2.2.1 Creating a SMACC file

There are two ways to make one:

- **From scratch, in the Editor.** In the Launcher, click **Editor…**. In the dialog, choose **New SMACC file** for a blank configuration (or pick an existing file to edit). Configure each panel on the left, set the data directory, and **Save SMACC file as…** to a new `.smacc` anywhere you like. The Editor never records a run — it only authors configuration.
- **From a live session.** In a running **Session**, use **File › Save SMACC file as…** to snapshot the session's *current* settings — including any mid-session tweaks (volumes, device routing, event-code edits) — to a new SMACC file. Handy when you've tuned things live and want tonight's setup to be tomorrow's starting point.

SMACC ships a `default.smacc` in your SMACC directory as a working example. It is treated as a read-only template — saving over it redirects to Save-As — so there is always a known-good configuration to fall back on.

You can keep SMACC files anywhere: for instance one per participant (`peter.smacc` , `paul.smacc` , …) or one per protocol, each pointing at whatever data directory you like (they can share one).

2.2.2 The data directory

Every SMACC file names a **data directory** — where its runs are written. Each session started from the file gets its own timestamped folder under that directory (e.g. `smacc-20260607-223015/`) holding the run's `.log` , dream-report recordings, survey responses, and any exports. Set it in the Editor (**Change data directory…**); choose it deliberately, since it determines where a night's data ends up.

2.2.3 Opening a SMACC file

Opening a SMACC file starts a new session configured by it:

- **Double-click the file** (Windows build): SMACC opens the **Start a session** dialog with that file preselected, so you confirm the subject/session and click **OK** to start. Enable or repair the file association any time from **File › Associate .smacc files (Windows)** in the Launcher.
- **From the Launcher**: click **Session…**, then pick the file in the **Start a session** dialog (recent files are listed; **Browse…** finds any other) and click **OK**.
- **From a terminal**: `SMACC path/to/file.smacc` .